

BWIN817 Seminar 4

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#DiscoverNWU

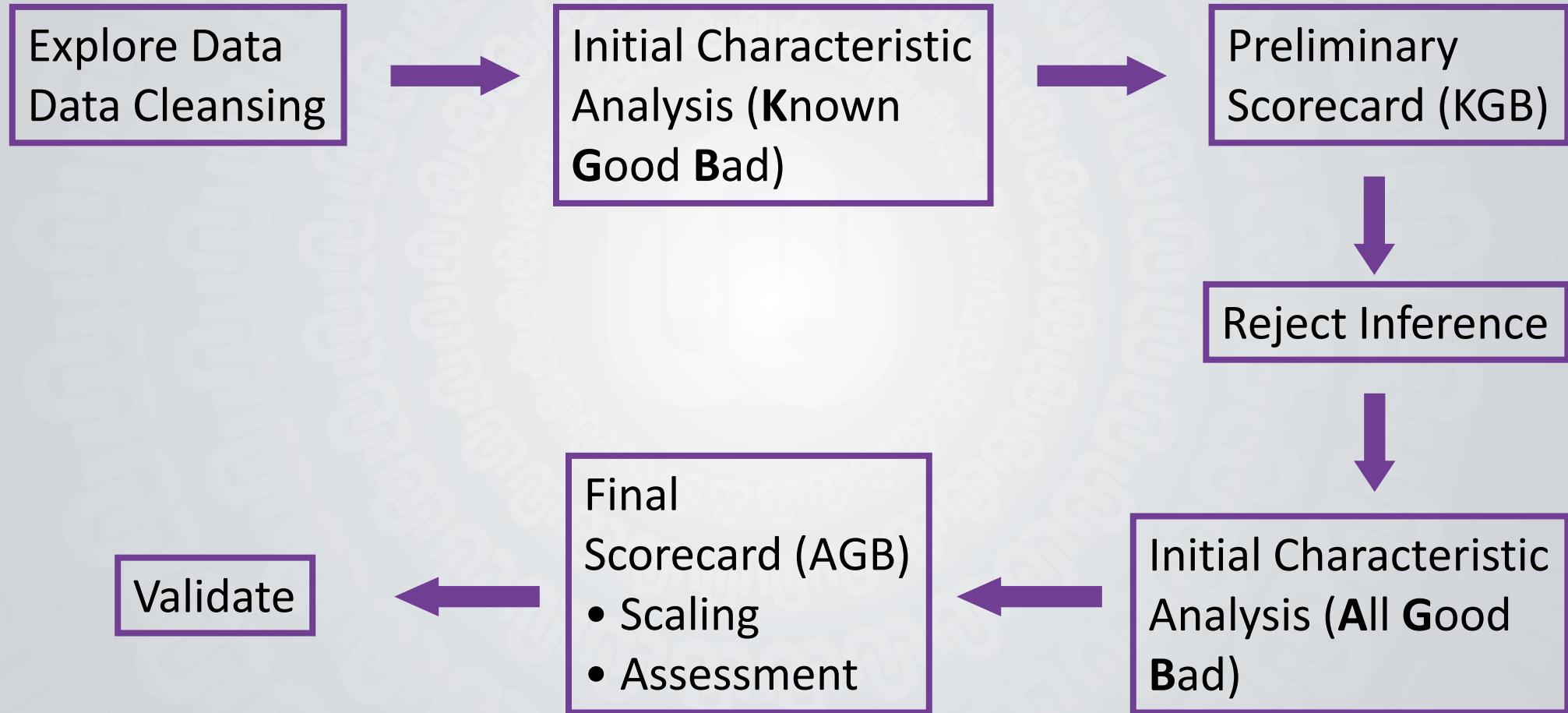
UWC module code:
COF827

Stage 4: Scorecard Development

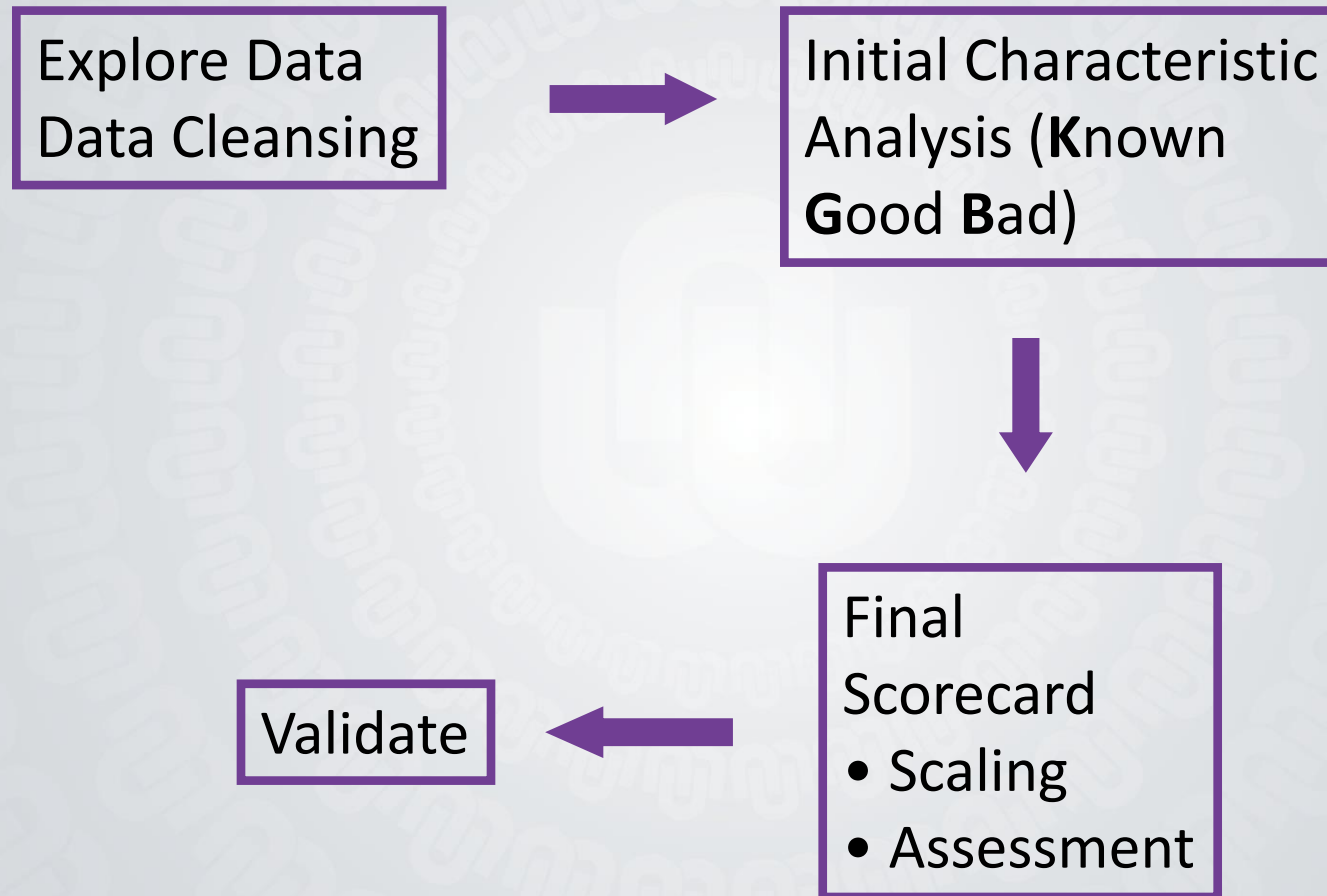
- ❖ Process flow
- ❖ Exploratory data analysis
- ❖ Missing values
- ❖ Outliers
- ❖ Initial characteristic analysis (grouping/classing)
- ❖ Scorecard (logistic regression)
- ❖ Reject inference
- ❖ Scorecard scaling
- ❖ Misclassification
- ❖ Scorecard strength
- ❖ Validation

Chapter 6 in Siddiqi

Process Flow – Application Scorecard



Process Flow – Behavioral Scorecard



Missing Values and Outliers

❖ **Missing**

- Exclude characteristics or records with significant missing values
- Group 'missing' as a distinct attribute
- Impute missing values

❖ **Outliers (and mis-keys)**

- Exclude records

Correlation

- ❖ Two methods to identify correlation in SAS:
 - ❖ PROC VARCLUS and PROC CORR
- ❖ Why is it important to analyze correlations?
 - ❖ Prevents the overemphasis of certain characteristics
 - ❖ Requirement of logistic regression
- ❖ Does multicollinearity create problems when dealing with predictive models?
 - ❖ It is not a problem when one develops models for predictive purposes with large datasets. The effects of multi-collinearity in reducing the statistical power of a model can be overcome by using a large enough sample such that the separate effects of each input can still be reliably estimated. (Siddiqi, 2006)
 - ❖ For predictive purposes “multicollinearity is not quite as damning” (Vaughan & Berry 2005)

Correlation and multi-collinearity

We considered the problem of multicollinearity that may arise from the inclusion of highly correlated independent variables (Dohoo et al., 1997). Therefore, correlations between all pairs of predictor variables were calculated. The highest correlations involve both the official and OTA “star ratings”. Thus, coefficients for two separate models (for three different samples available at different OTAs) were estimated: one involving the official “star rating” as a predictor of hotel room rates, and another one involving OTA “star rating”. **It is argued that multicollinearity is certain at the 0.9 level of a correlation coefficient or higher (Dohoo et al., 1997).** However, no certain rules exist for the determination of significance of the multicollinearity phenomenon in a particular regression analysis application (Chen and Rothschild, 2010). Furthermore, we employed the VIF statistic to verify the existence of multicollinearity (Hung et al., 2010). **The VIF values for all variables are below the critical value – equal to 10 as Lin (2008) has suggested.**

References:

Dohoo, I., Ducrot, C., Fourichon, C., Donald, A. and Hurnik, D. (1997), “An overview of techniques for dealing with large numbers of independent variables in epidemiologic studies”, Preventive Veterinary Medicine, Vol. 29 No. 3, pp. 221-239.

Lin, F.J. (2008), “Solving multicollinearity in the process of fitting regression model using the nested estimate procedure”, Quality and Quantity, Vol. 42 No. 3, pp. 417-426

Rules of thumb in industry:

Correlation 0.6 & VIF 4

Note: rule of thumb means broad guideline, not set in stone

Some techniques are more sensitive to multicollinearity (e.g. linear regression), while others are less affected (e.g. random forests, decision trees).

Even in less sensitive methods, removing highly correlated variables is still valuable because:

- They add redundant information.
- They increase model complexity and reduce interpretability.
- They may inflate the variance of coefficient estimates (in regression).
- They contribute unnecessary degrees of freedom.
- They can negatively impact feature importance ranking or variable selection.
- They can cause instability in model performance across different samples

Initial Characteristic Analysis (classing/grouping/binning)

- ❖ Identify strong characteristics
 - Best differentiators between 'good' and 'bad'
 - Screening
- ❖ Group the attributes
 - Scorecard format

Binning

The concept of binning is known by many names such as discretisation, classing, categorisation, grouping and quantisation. Binning is the mapping of continuous or categorical data into discrete bins (Nguyen, Müller, Vreeken and Böhn, 2014). It is an important pre-processing step in most predictive models, and considered a basic data preparation step in building a credit scorecard (Thomas, 2009).

Criteria for Selection

- ❖ Statistical measures
 - Weight of Evidence
 - Information Value
 - Chi Square
- ❖ Logical trend for “weight of evidence”
- ❖ Operational/business consideration

Weight of Evidence

Age	Count	Distr Count	Goods	Distr Good	Bads	Distr Bad	Weight of Evidence	Information Value Calculation
Missing	51	2.54%	43	2.367%	8	4.124%	-0.555325335	0.009758
18-22	200	9.95%	152	8.365%	48	24.742%	-1.084404399	0.177591
23-26	300	14.92%	246	13.539%	54	27.835%	-0.72073642	0.103038
27-29	450	22.38%	405	22.289%	45	23.196%	-0.039859332	0.000361
30-35	500	24.86%	475	26.142%	25	12.887%	0.70735507	0.093763
35-44	360	17.90%	349	19.207%	11	5.670%	1.22009274	0.165169
44 +	150	7.46%	147	8.090%	3	1.546%	1.654736389	0.108284
Total	2,011		1,817		194			0.6580
Information Value = 0.6580								

Distr. Good = # goods (attribute)/ # total goods=43/1817=2.367%

Weight of evidence (WOE)= LN (Distr.Good/Distr. Bad) = LN (2.367%/4.124%)=-0.55532

Information Value = $\sum(\text{Distr. Good} - \text{Distr. Bad}) * \text{WOE}$

$$IV = \sum \left((\%Good - \%Bad) \times \ln \frac{\%Good}{\%Bad} \right)$$

**Open Excel file:
weight of evidence.xls**

Need to memorise these formulas. Need to be able to this on paper (for theory tests/exams) and on computer (for practical tests/exams)

Notation of WOE and IV

The number of events (n_j^{event}) and the number of nonevents ($n_j^{nonevent}$) in bin j

N^{event} and $N^{nonevent}$ are the total numbers of events and non-events in the data

The WOE is calculated for each bin j (of every variable) as follows:

$$WOE_j = \ln \left(\frac{n_j^{nonevent}}{N^{nonevent}} / \frac{n_j^{event}}{N^{event}} \right)$$

The IV for each variable i is defined as follow: $IV_i = \sum_{j=1}^{J_i} \left(\frac{n_j^{nonevent}}{N^{nonevent}} - \frac{n_j^{event}}{N^{event}} \right) \times \ln \left(\frac{n_j^{nonevent}}{N^{nonevent}} / \frac{n_j^{event}}{N^{event}} \right)$

$$\text{Alternatively, } IV = \sum \left((\%Good - \%Bad) \times \ln \frac{\%Good}{\%Bad} \right)$$

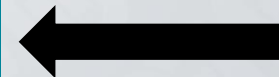
Weight of Evidence & Information Value

WOE

- ❖ Measures strength of each binned attribute (variable) in separating goods and bads
- ❖ $\text{Distr Good} / \text{Distr Bad} = \text{odds of being good}$
- ❖ Negative weight : more bads than goods
- ❖ Logical trend

Rule of thumb of IV (differ from company to company)

Information Value	Variable Predictiveness
Less than 0.02	Not useful for prediction
0.02 to 0.1	Weak predictive Power
0.1 to 0.3	Medium predictive Power
0.3 to 0.5	Strong predictive Power
>0.5	Suspicious Predictive Power



A high Information Value (IV) can arise usually due to one of 2 reasons:

- **Reason 1: Target Leakage:** When a feature inadvertently includes information from the target variable, it can result in a misleadingly high IV, overestimating predictive power.

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- **Reason 2: Strong Predictor:** Some features genuinely have strong predictive power, especially if they are closely related to the outcome.

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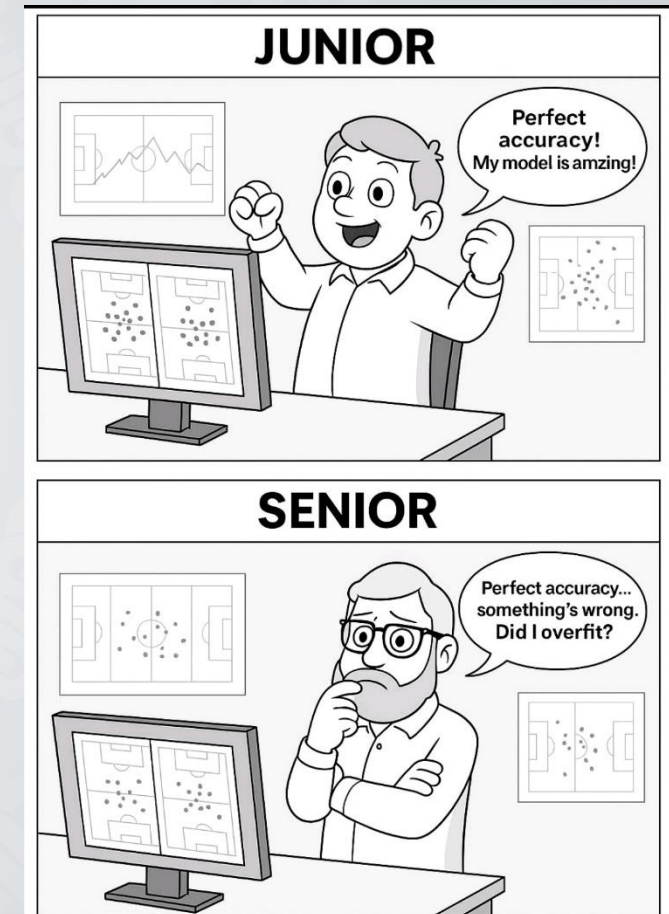
Careful analysis is required to distinguish between these two scenarios, often through subject-matter expertise and review of data collection processes.

This also applies when metrics like the Gini coefficient, ROC AUC, or accuracy are **too high**. If something seems too good to be true, it probably is.

See this LinkedIn meme:

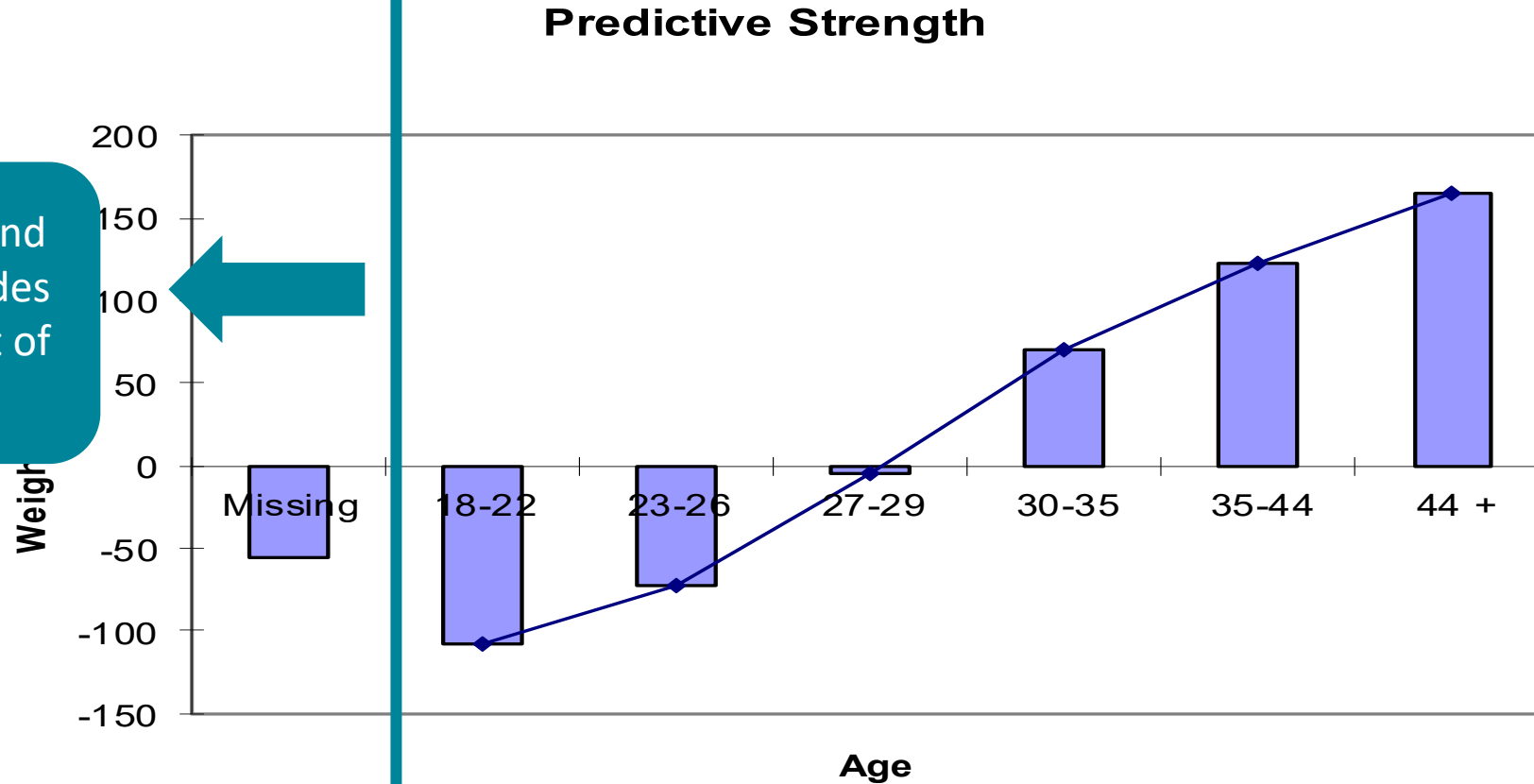
Junior analyst: *Perfect accuracy! My model is amazing!*

Senior analyst: *Perfect accuracy? Something's wrong. Did I overfit?*



Logical Trend

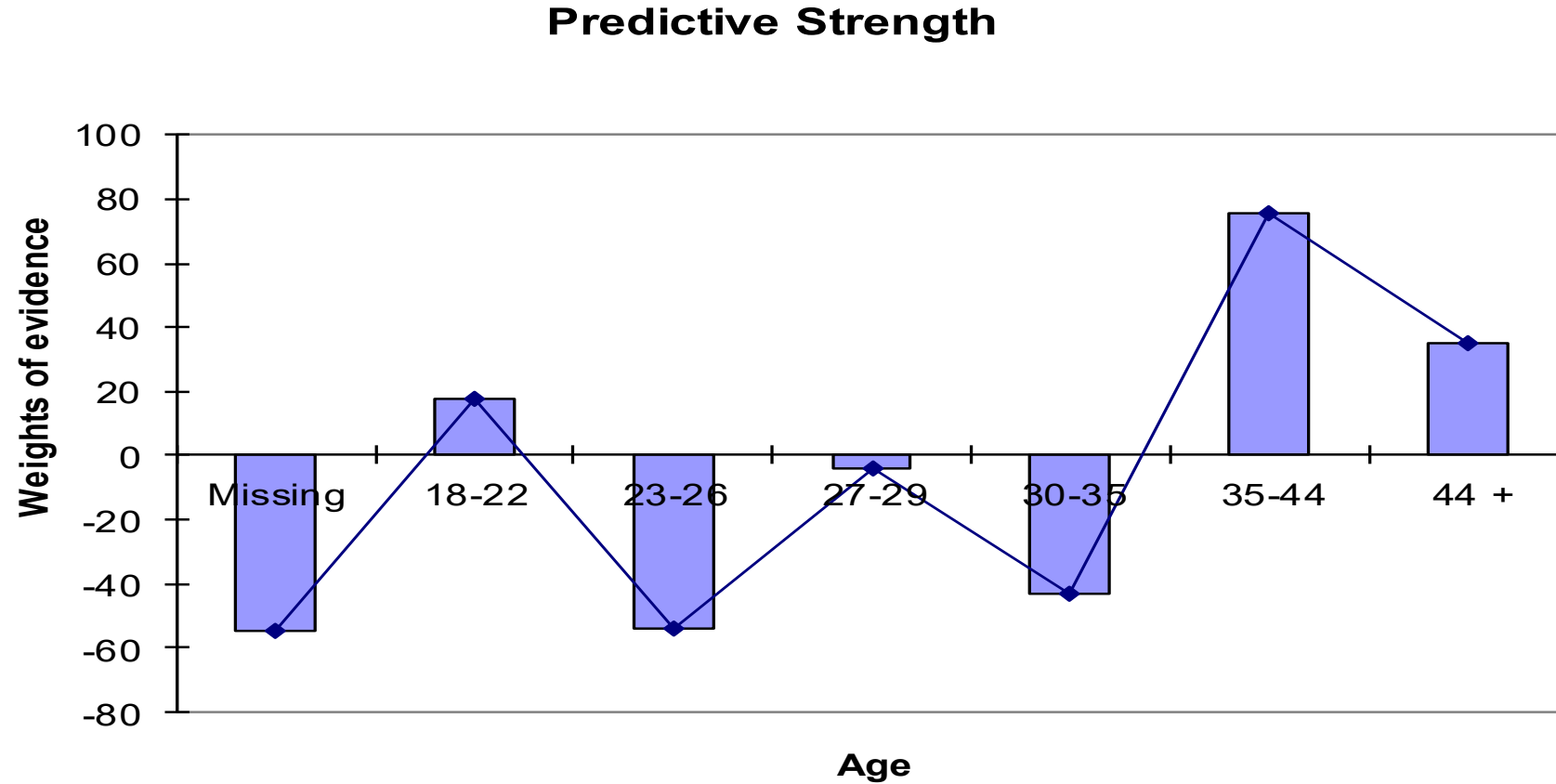
Missing values and other special codes do not form part of logical trend



Logical Trend

- ❖ Final weightings make sense
- ❖ Buy-in
- ❖ Confirms business experience
 - young people are higher risk
- ❖ Reduces overfitting if done correctly - model overall trend, not quirks

Logical Trend

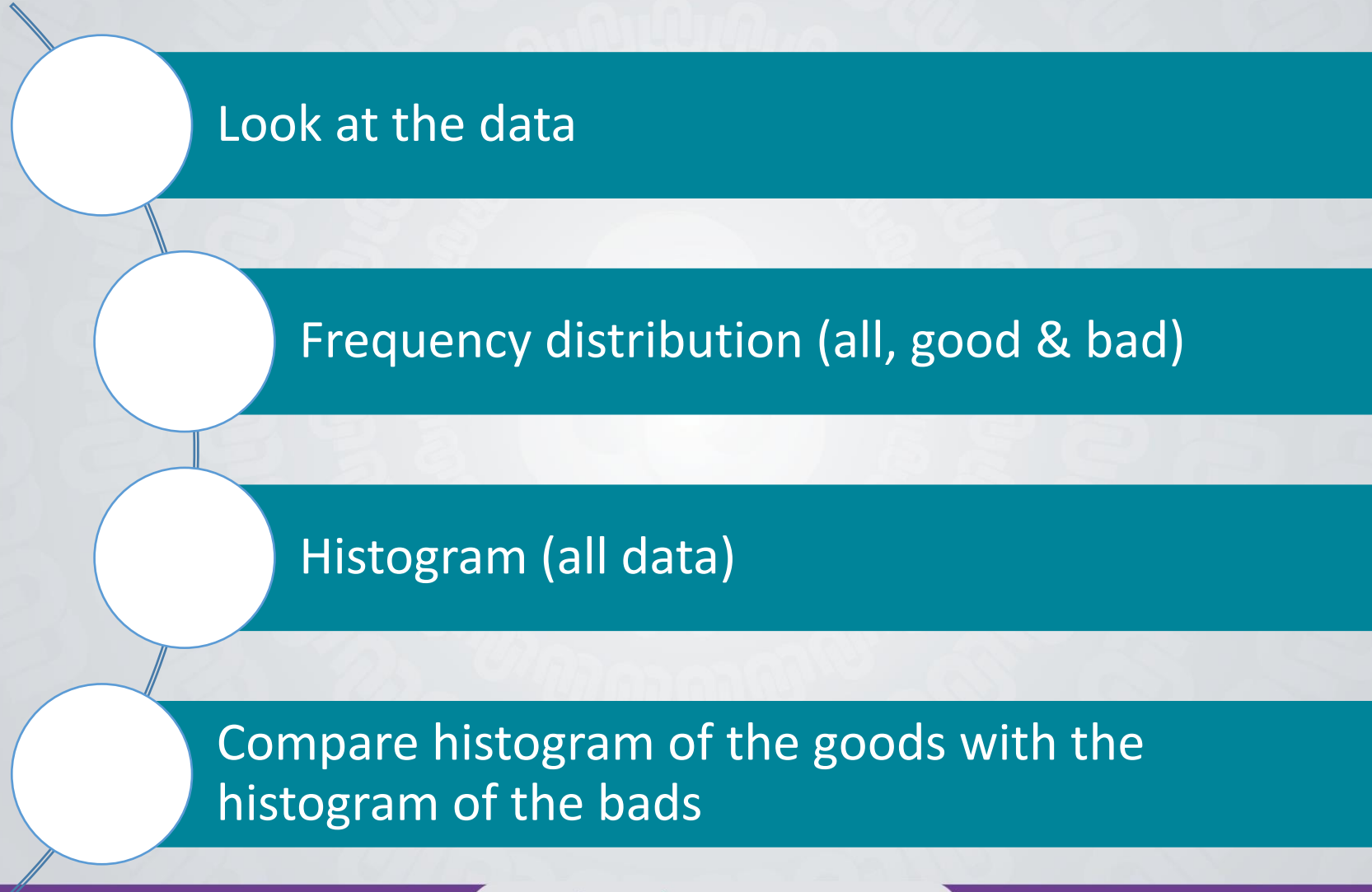


Binning, WOE, IV

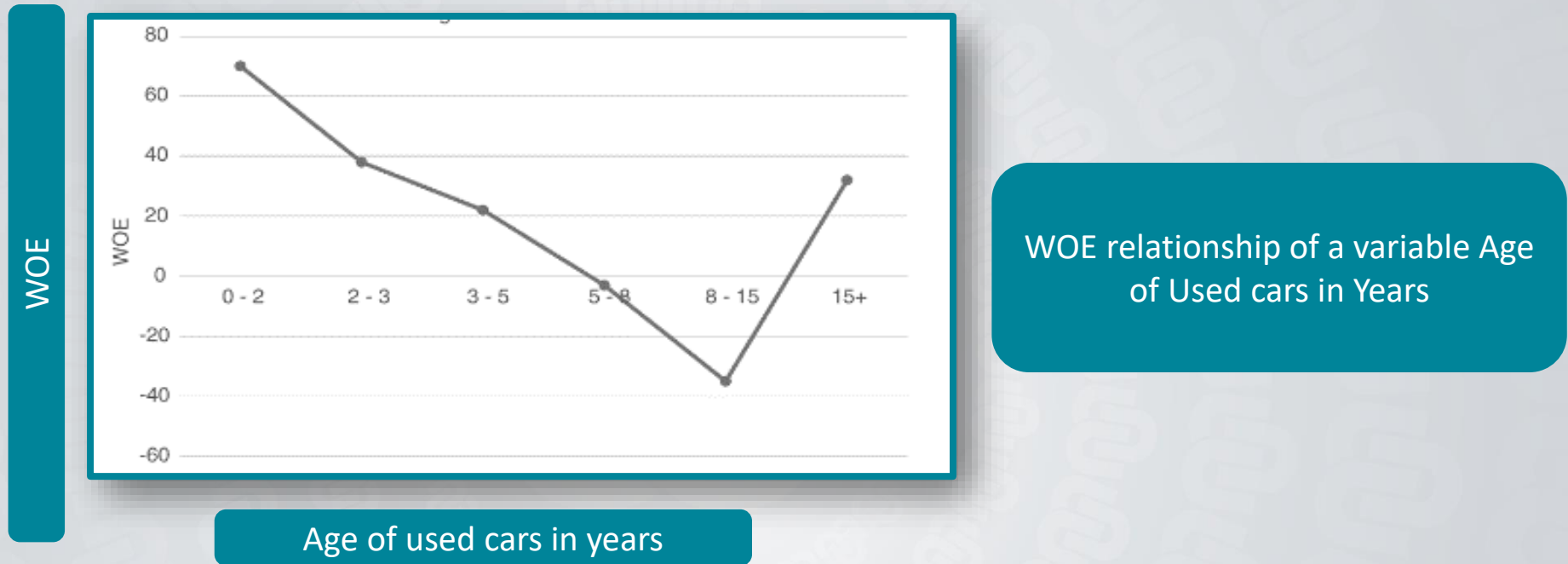
TIP

Instead of striving for bins with roughly equal numbers of observations, your goal should be to create bins that effectively separate between positive and negative outcomes.

How to start binning?



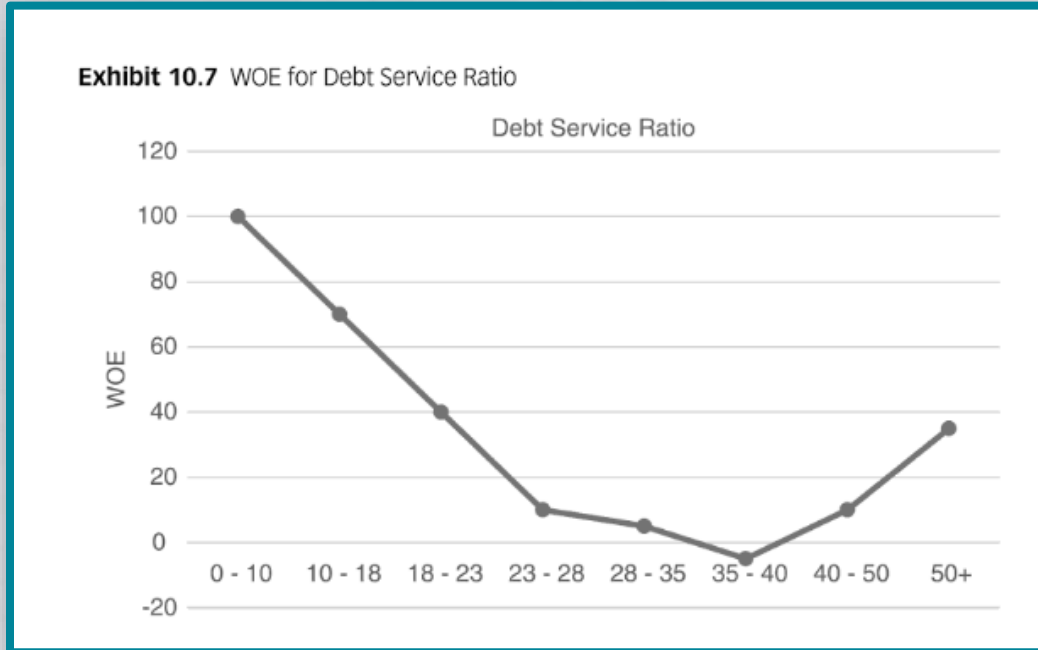
Example : Auto Loan - Age of Used cars in Years



- ❖ WOE relationship for a variable “Age of Used cars in Years” for an auto loans scorecard (Page 187 in Siddiqi, 2017)
- ❖ The chart shows a trend where generally, the older the used car the borrower is buying, the worse their performance. This seems to make sense, except the last bucket, where people buying cars older than 15 years old perform better than expected
- ❖ This should be confirmed with business. The data in 15+ might be correct and represents people who are buying expensive vintage cars. These would tend to be richer people, who presumably are better at paying their loans

Example : Auto Loan – Debt Service Ratio

WOE



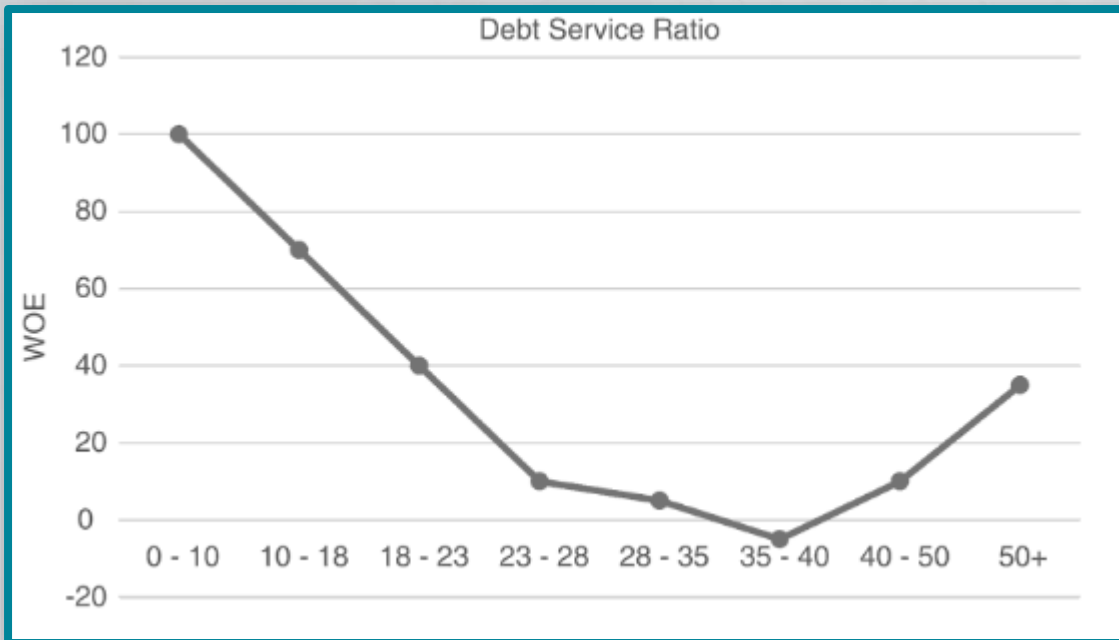
Debt Service Ratio

WOE relationship of a variable
Debt Service Ratio

DSR: 23% if Income R10 000 and Debt: R2300

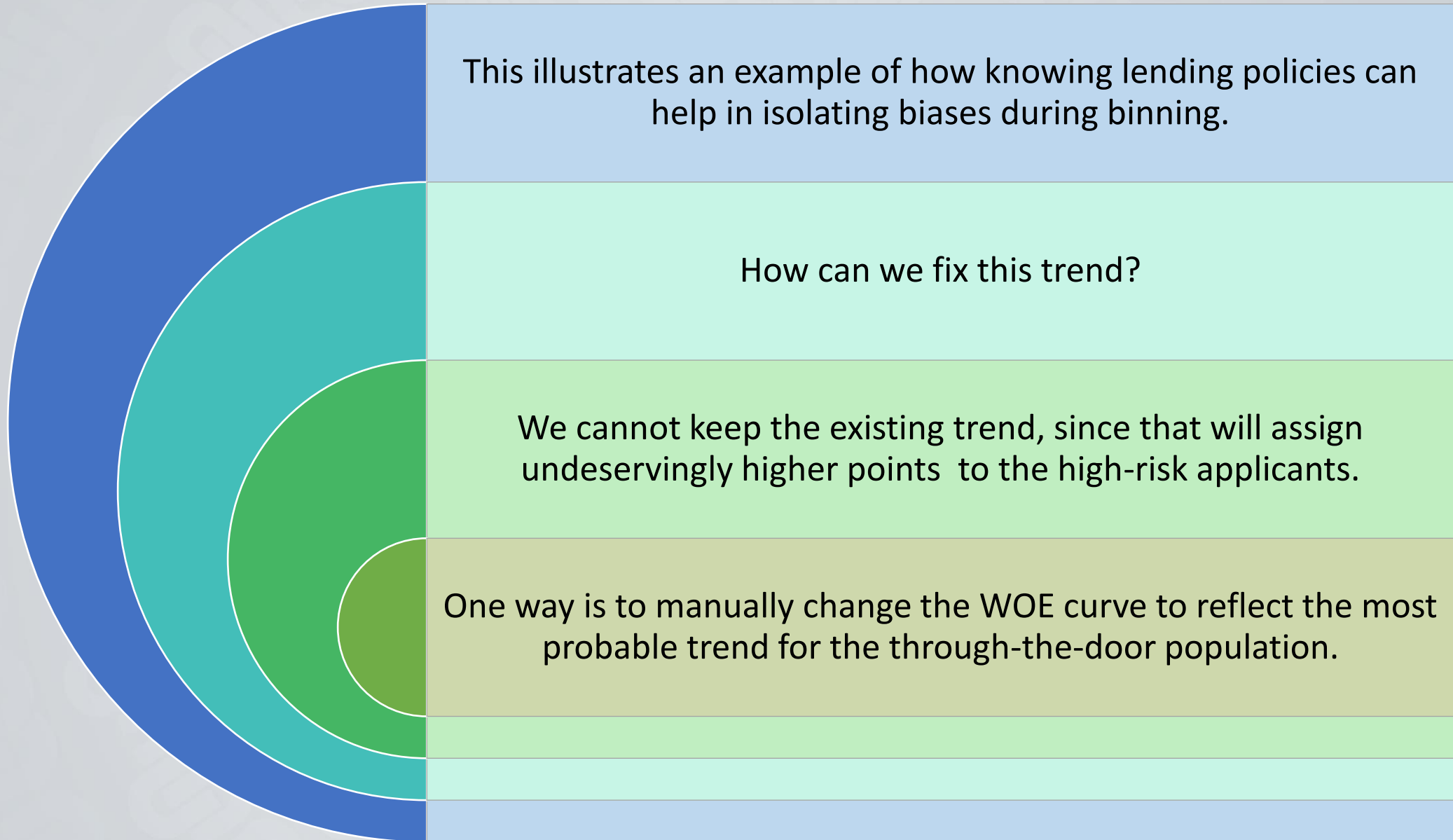
(Debt consist for example of R1200 home loan,
Car loan R800, R300 for credit cards)

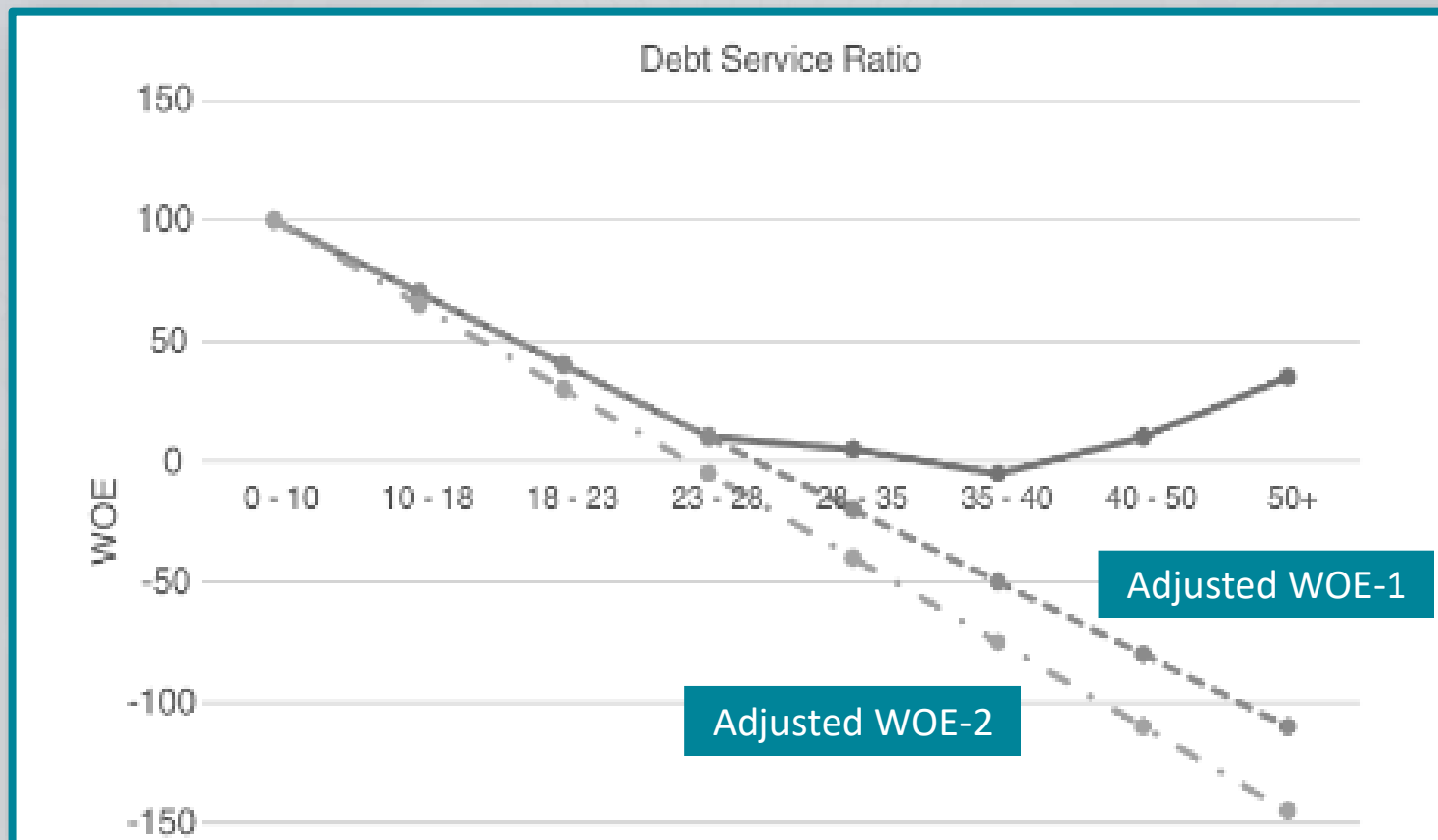
- ❖ WOE relationship for a variable “WOE for Debt Service Ratio” for an auto loans scorecard (Page 189 in Siddiqi, 2017, Exhibit 10.7)
- ❖ Another way variables are biased is through policy rules. Look at Debt Service Ratio.
- ❖ This chart shows the WOE relationship for debt service ratio (DSR), which is generally defined as the percentage of income going towards servicing debt



- ❖ Fairly linear from 0 to 28
- ❖ 28 to 40 trend still logical, but gradient changed
- ❖ Beyond 40, reversal in logic

- ❖ Reverse in logic, **since experience and business sense dictates that the higher the DSR, the worse the performance.**
- ❖ What happened here? **Knowledge of policies will answer this!**
- ❖ The bank follows normal lending policies up to 28% DSR
- ❖ From 28 to 40 %, due to elevated risks, the bank has additional policy rules (based on higher bureau scores, higher income levels. This zone has performance biased by policy rules.
- ❖ Above 40 percent, the lending policy allowed only manual approvals by experienced adjudicators. This is the most biased zone.





- ❖ Exhibit 10.8 in Siddiqi (2017) shows two examples of how the original WOE can be adjusted
- ❖ The first line “Adjusted WOE-1” is based on an extrapolation of the part of the chart with normal lending policies
- ❖ The second line, “Adjusted WOE-2” assumes a worse performance

Remember Gini? Somers' D?

C-statistic	Gini
AUROC	Wilcoxon Mann-Whitney U statistic
Area under the ROC curve	Summary statistic under CAP curve
c	Somer's D
ROC index	Accuracy ratio
Range: 0.5-1	Range: 0-1
Perfect AUROC: 1	Perfect Gini: 1
Random AUROC: 0.5	Random Gini: 0

Let's focus on ROC index

$$Gini = 2(c) - 1$$

Confusion Matrix

The **confusion matrix** is a cross-tabulation of the actual and predicted classes. It quantifies the confusion of the classifier.

Simplest performance statistics:

Accuracy = (true positives and negatives) / (total cases)

Error Rate = (false positives and negatives) / (total cases)

		Predicted Class		
		0	1	
Actual Class	0	True Negative	False Positive	Actual Negative
	1	False Negative	True Positive	Actual Positive
		Predicted Negative	Predicted Positive	

Confusion matrix



Sensitivity and Positive Predicted Value

Sensitivity = (true positives) / (total actual positives)

Positive predicted value (PV+) = (true positives) / (total predicted positives).

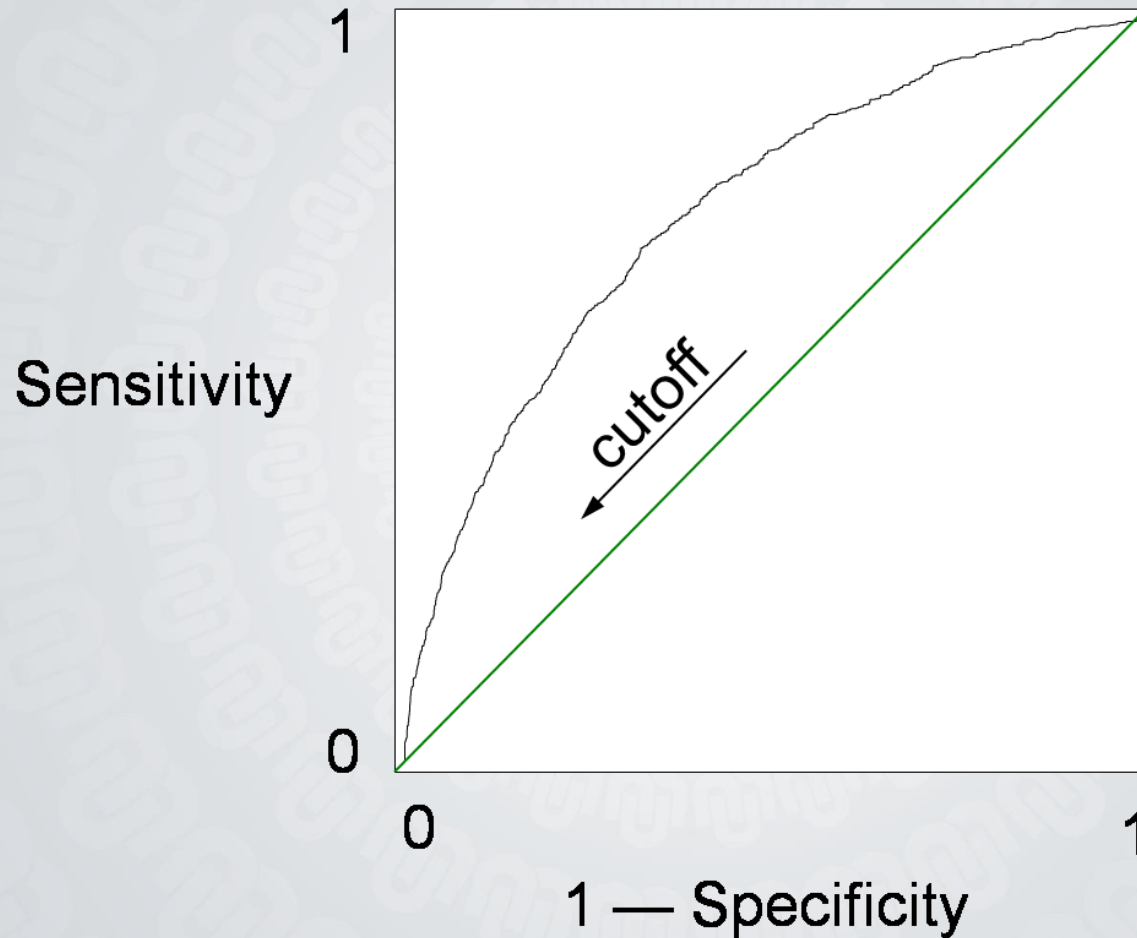
Specificity = (true negatives) / (total actual negatives)

Negative predicted value (PV-) = (true negatives) / (total predicted negatives).

		Predicted Class		
		0	1	
Actual Class	0			Actual Positive
	1		True Positive	
		Predicted Positive		

ROC Curve

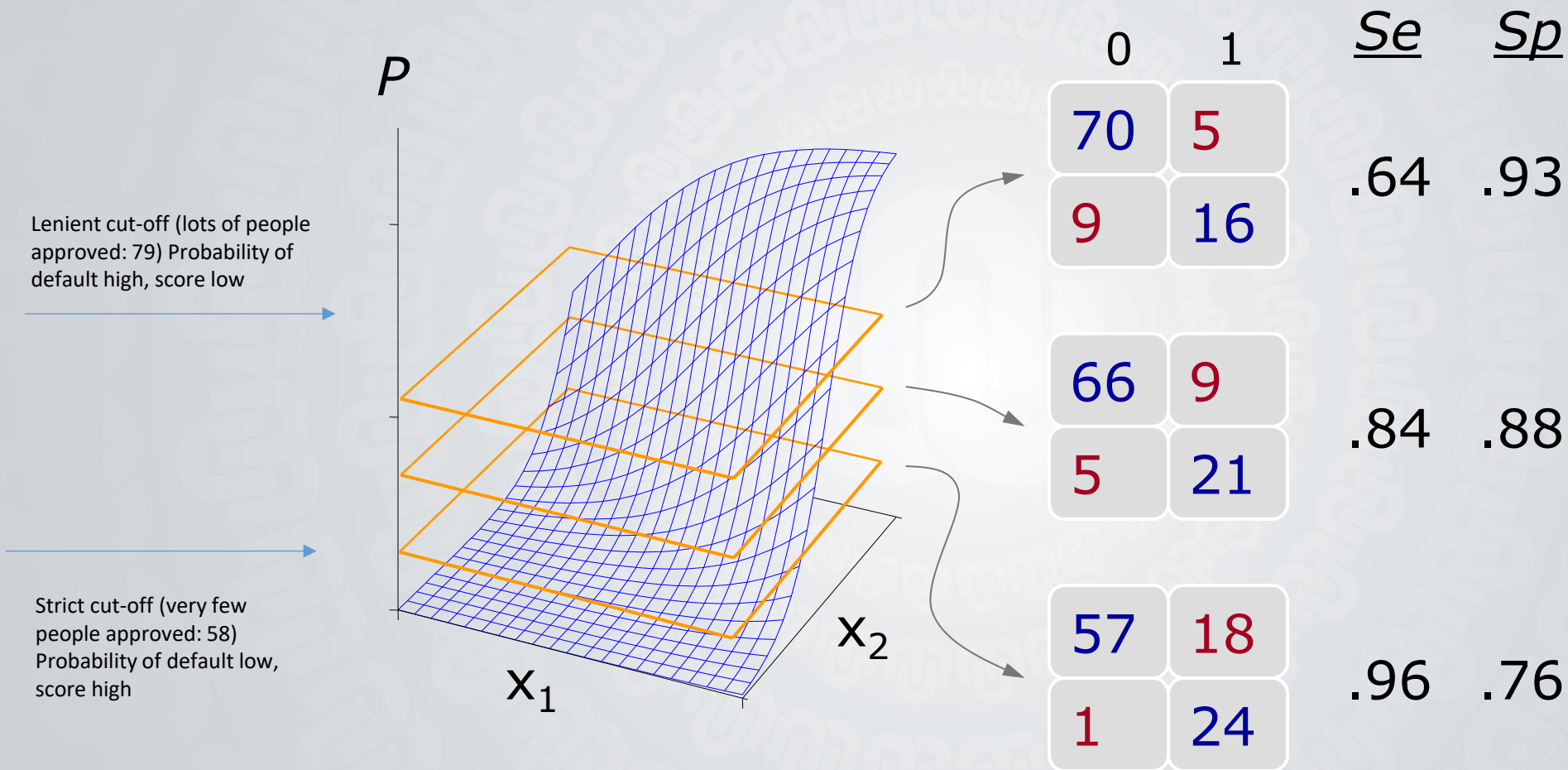
As the cutoff decreases, more and more cases are allocated to class 1; hence, the sensitivity increases and specificity decreases.



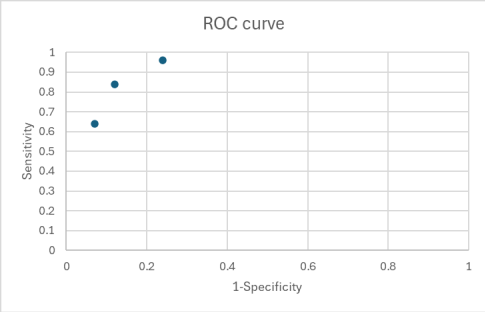
The **receiver operating characteristic (ROC) curve** was adapted from signal detection theory for the assessment of classifiers.

Cutoffs

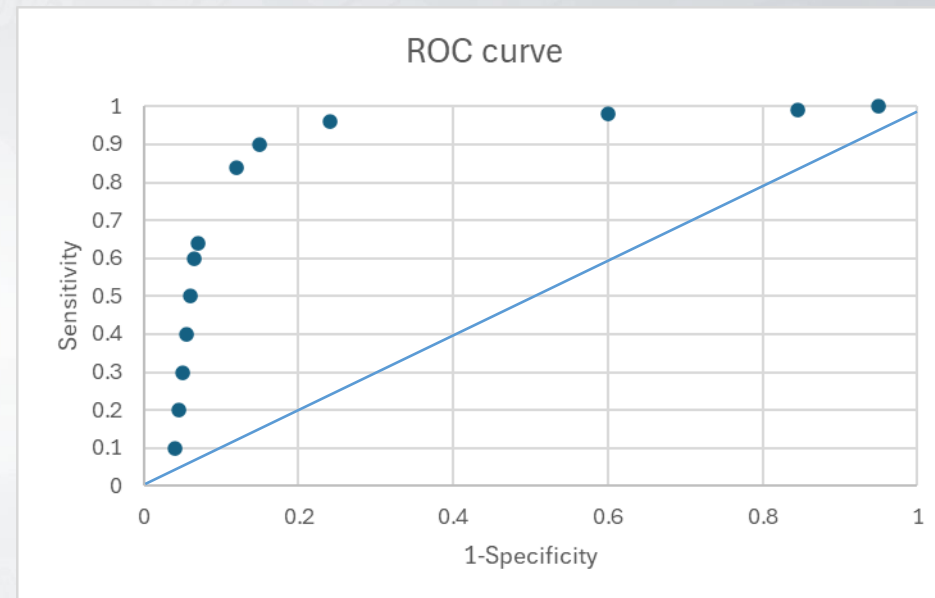
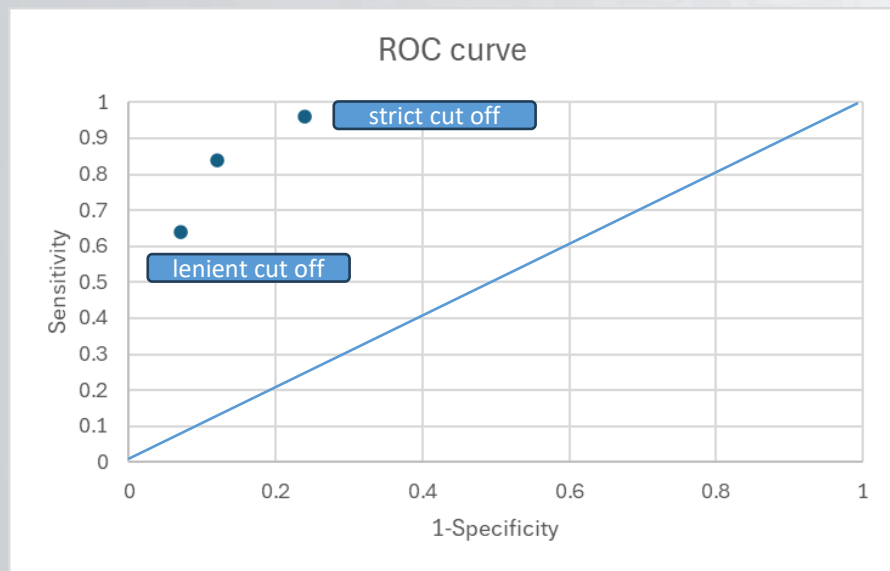
3 different cutoffs result in 3 different confusion matrices



Specificity	1- Specificity	Sensitivity
0.93	0.07	0.64
0.88	0.12	0.84
0.76	0.24	0.96



Specificity	1- Specificity	Sensitivity	# of approved	Cut off
0.93	0.07	0.64	70+9 = 79	low cut off
0.88	0.12	0.84	66+5 = 71	medium cut off
0.76	0.24	0.96	57+1 = 58	high cut off (strict)



Oversampled Test Set

		Predicted		
		0	1	
Actual	0	29	21	50
	1	17	33	50
		46	54	
		Sample		

		Predicted		
		0	1	
Actual	0	56	41	97
	1	1	2	3
		57	43	
		Population		

Adjustments for Oversampling

		Predicted Class		
		0	1	
Actual Class	0	$\pi_0 \cdot Sp$	$\pi_0(1 - Sp)$	π_0
	1	$\pi_1(1 - Se)$	$\pi_1 \cdot Se$	π_1

This is the formula to adjust!

Knowing sensitivity, specificity, and the priors is sufficient for adjusting the confusion matrix for oversampling

Adjustments for Oversampling

		Predicted Class		
		0	1	
Actual Class	0	$\pi_0 \cdot Sp \cdot N$	$\pi_0(1 - Sp) \cdot N$	π_0
	1	$\pi_1(1 - Se) \cdot N$	$\pi_1 \cdot Se \cdot N$	π_1

To get the corrected matrix, these formulas must be multiplied with **N** (the total number of observations in the dataset)

Knowing sensitivity, specificity, and the priors is sufficient for adjusting the confusion matrix for oversampling

Complete Assignment L11 (in class)

Tips for Assignment L11
Tips how to bin (Page 77 to 87 Siddiqi)

- Missing is grouped separately (if there is enough observations – guideline at least 2% of observations)
- A general “minimum 5% in each bucket” (rule of thumb)
- No groups with 0 counts for good or bad
- Logical trend for WOE very IMPORTANT (give a sentence or two to explain the logical trend)
- Information value important
- Not too many observations in one bin (rule of thumb: not more than 50%)
- Initial binning might be done with Decision Trees (SAS EM) or Interactive Grouping Node (SAS EM) or PROC HPBIN or PROC SPLIT
- More than one possible solution

What is your goal with binning

(in a test or in the exam or in an assignment or in BWIR828 – industry project second semester)?

- Your goal should be to create bins that effectively separate between positive and negative outcomes.
- You must choose a binning that is optimal
How will you measure optimality?
Information value and WOE logical trend can be used
- Usually you start with a binning, check the IV (information value) and logical trend. Try a new binning, check again the IV, and Logical trend. After a few iterations, you are satisfied that you have the best binning.
- OR you can get this "best" binning quicker by starting with a good first guess! How can you get a good first guess? At the bank, the previous' scorecard's binning, might be a good place to start.

Tips for Assignment L11: Note on SAS code

DemonstrationsSeminar4 a _g

(d)

```
proc summary data=tt nway;
```

```
class class;
```

```
var LOAN;
```

```
output out = tabel1 sum=sum;
```

```
run;
```

```
proc summary data=tt nway;
```

```
class class bad;
```

```
var loan;
```

```
output out = tabel2 sum=sum;
```

```
run;
```

Class: The binned DEROG variable

LOAN: **any** numerical variable can be used here, so that proc summary can work. You can even just add a new column with the value 1 for all observations



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by Mark Parisi

I WAS ONLY
GONE FOR A
MOMENT!

FOR ME
IT'S **SEVEN**
MOMENTS!



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